Spectral Adversarial Feature Learning for Anomaly Detection in Hyperspectral Imagery

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***Abstract*— Theoretically, hyperspectral images (HSIs) are capable of providing subtle spectral differences between different materials, but in fact, it is difficult to distinguish between background and anomalies because the samples of anomalous pixels in HSIs are limited and susceptible to background and noise. To explore the discriminant features, a spectral adversarial feature learning (SAFL) architecture is specially designed for hyperspectral anomaly detection in this article. In addition to reconstruction loss, SAFL also introduces spectral constraint loss and adversarial loss in the network with batch normalization to extract the intrinsic spectral features in deep latent space. To further reduce the false alarm rate, we present an iterative optimization approach by a weighted suppression function that depends on the contribution rate of each feature to the detection. In particular, the structure tensor matrix is adopted to adaptively calculate the contribution rate of each feature. Benefiting from these improvements, the proposed method is superior to the typical and state-of-the-art methods either in detection probability or false alarm rate.**

***Index Terms*— Adversarial learning, anomaly detection, feature extraction, hyperspectral image (HSI), iterative optimization.**

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I. Introduction

HYPERSPECTRAL images (HSIs) have three dimensions, one of which collects abundant spectral information of surface materials, and the other two dimensions contain spatial information of pixels. Based on such capabilities, HSIs are widely used in various fields, such as land cover classification [1]–[3], target detection [4], [5], and anomaly detection [6]–[8]. Especially, anomaly detection is more attractive in military and civilian applications because of the limited samples and lack of prior knowledge. In an HSI, a so-called anomaly refers to a pixel whose spectral and spatial features are significantly different compared to the background [9]. Different from the supervised target detection, anomaly detection is based on the unsupervised or semisupervised methods without prior knowledge of the target or background, so it is more conducive to exploring unknown environments or targets, such as deep space exploration and new fighters monitoring [10].

The traditional methods utilize the background modeling or statistical distributions between background and anomalies to determine anomalous pixels. The algorithm proposed by Reed and Yu (RX) [11] is a well-known anomaly detection method, which can be viewed as a milestone approach because the assumption that the background obeys the multivariate normal distribution is applied to anomaly detection for the first time. The essence of RX is to locate anomalies by computing the Mahalanobis distance between the test pixel and the background. However, the estimation of the mean vector and covariance matrix describing the background may be inaccurate because of the mixing of anomalies and noise. Furthermore, complex backgrounds in real HSIs cannot be accurately described as simple Gaussian distributions in practical applications. Therefore, numerous improved methods based on RX have continuously been proposed. Among them, the local RX detector (LRX) [12] describes background using local statistics, whereas the global RX (GRX) [12] utilizes global statistics to do so. The kernel RX [13] aims to model the background in a higher dimensional space by kernel functions. However, it is challenging to select an appropriate kernel function, and the complexity of general kernel functions leads to the enormous computational cost [13]–[15].

Moreover, non-RX-based methods are also the focus of research. The collaborative representation-based detector (CRD) [16] assumes that a pixel in the background can be

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approximately linear combined by its spatial neighborhoods, whereas anomalies cannot, and the weight vector is minimized by reinforcing the collaboration of representation. Under the hypothesis that anomalies are sparse while the background is low-rank in HSIs, Xu *et al.* [17] separated the background and anomalies in the observed data by adopting a background dictionary and representation coefficients of all pixels to model the background part and then applying the response of the residual matrix. In addition to the methods mentioned earlier, recently, deep learning-based methods have attracted more attention. Sun *et al.* [18] proposed a randomized subspace learning-based anomaly detector (RSLAD) for HSIs. For extracting effective feature of HSIs, a novel anomaly detection method via discriminative feature learning with multiple-dictionary sparse representation was proposed in [19]. Li *et al.* [9] first trained a deep convolutional neural network (CNN) for hyperspectral anomaly detection. Besides, a general end-to-end 2-D CNN (GETNET) framework was proposed in [20] for HSI change detection (HSI-CD). Zhao *et al.* [10] proposed an anomaly detection method based on an improved autoencoder (AE) called stacked denoising AE (SDAE). Besides, as an extension of AE, the adversarial AE (AAE) proposed in [21] is also developed for unsupervised detection of lesions in brain MRI [22] and anomaly detection in videos [23]. The AAE-based methods have a leading performance for anomaly detection in different fields. However, in hyperspectral anomaly detection, AAE has not yet been involved and developed. The main reason is that anomalies are rarely presented and their spectra are susceptible to the influence of surrounding pixels. Besides, it is complex to deal with high-dimensional redundancy information.

To address these problems, we establish a novel unsupervised network architecture specially designed for hyperspectral anomaly detection. First, a spectral adversarial feature learning (SAFL) model is presented to extract distinctive features in deep latent space. The quality of the deep spectral features is evaluated by the reconstruction error between the input spectral vector and the reconstructed one. The smaller the reconstruction error, the more powerful the ability of the extracted features to characterize the input spectral vector. Then, we conduct the averaging fusion method (AVG) on the extracted deep features to obtain a comprehensive feature. The morphological attribute filter (AF) [24] is utilized to enhance the response of anomalies because of their sparsity. To further reduce the false alarm rate, we introduce an adaptive weighted suppression function by calculating the contribution of each feature to anomaly detection. In particular, the suppression process is iterated until the detection result remains stable. Here, the structural similarity of two successive detection results is utilized as a criterion for stopping the detection.

In contrast with other anomaly detection algorithms, the main contributions of our newly proposed approach are summarized as follows.

1) A novel SAFL architecture is specially designed to extract features that keep the essential profiles of the input spectral vector to the maximum extent. In comparison to AE and AAE, SAFL is more beneficial to hyperspectral anomaly detection.
2) We introduce structure tensor (ST) matrix to adaptively calculate the contribution rate of each feature to anomaly detection.
3) We present an iterative detection approach by a suppression function that adopts the contribution of each feature. Extensive experiments on 15 HSIs demonstrate that our method can effectively detect anomalies while reducing the false alarm rate.

The remainder of this article is organized as follows. Section II is a brief introduction to AE and AAE. Section III presents the proposed method in detail. Section IV provides the data sets employed in the experiments for evaluation, parameter analysis, and experimental results. Finally, Section V concludes this article.

II. Related Work

A. AE

AE is an unsupervised neural network architecture that maps given samples x to an output image $\hat{x}$. It consists of an encoder function $f : x \to z$ that transforms input samples to hidden code z and a decoder function $g : z \to \hat{x}$ that attempts to restore the original input samples as precise as possible by using the hidden code z. Both the encoder and the decoder are composed of two fully connected layers and leverage sigmoid function as the nonlinear activation function. Besides, the hidden code z is regarded as a low-dimensional latent feature representation of the input samples that are equivalent to differentiating nonredundant information in the inputs.

Based on the fact that AE only focuses on the reconstruction error between input and reconstruction, it is trained to minimize the reconstruction error or rather the Euclidean distance between x and $\hat{x}$, which can be described as the function: $L = \|x - \hat{x}\|_2$. Once the training is terminated, the hidden code or the trained model can be adopted into anomaly detection [25] or classification [26].

B. AAE

AAE is one of the variants of AE, and it utilizes an adversarial training criterion to impose a prior on the latent space in contrast with AE. To do so, the discriminator is trained to discriminatively predict whether the samples are obtained from the hidden code or sampled from the prior distribution. In other words, AAE adds a discriminator network consisting of two fully connected layers between the encoder and the decoder. AAE is trained in two phases. First, the encoder and decoder constitute an AE to minimize the reconstruction error. Then, the encoder and discriminator make up a generative adversarial network (GAN) [27], of which the encoder is regarded as a generator. The discriminator is updated to distinguish the given samples from the generated samples obtained by the generator, and the encoder, i.e., generator, is updated to fool the discriminator.

AAE has been successfully applied in RGB or gray images [22], [23], [28]. Because of its powerful representational capability, we apply and develop it to dig intrinsic spectral features from high-dimensional spectral vectors. In particular,

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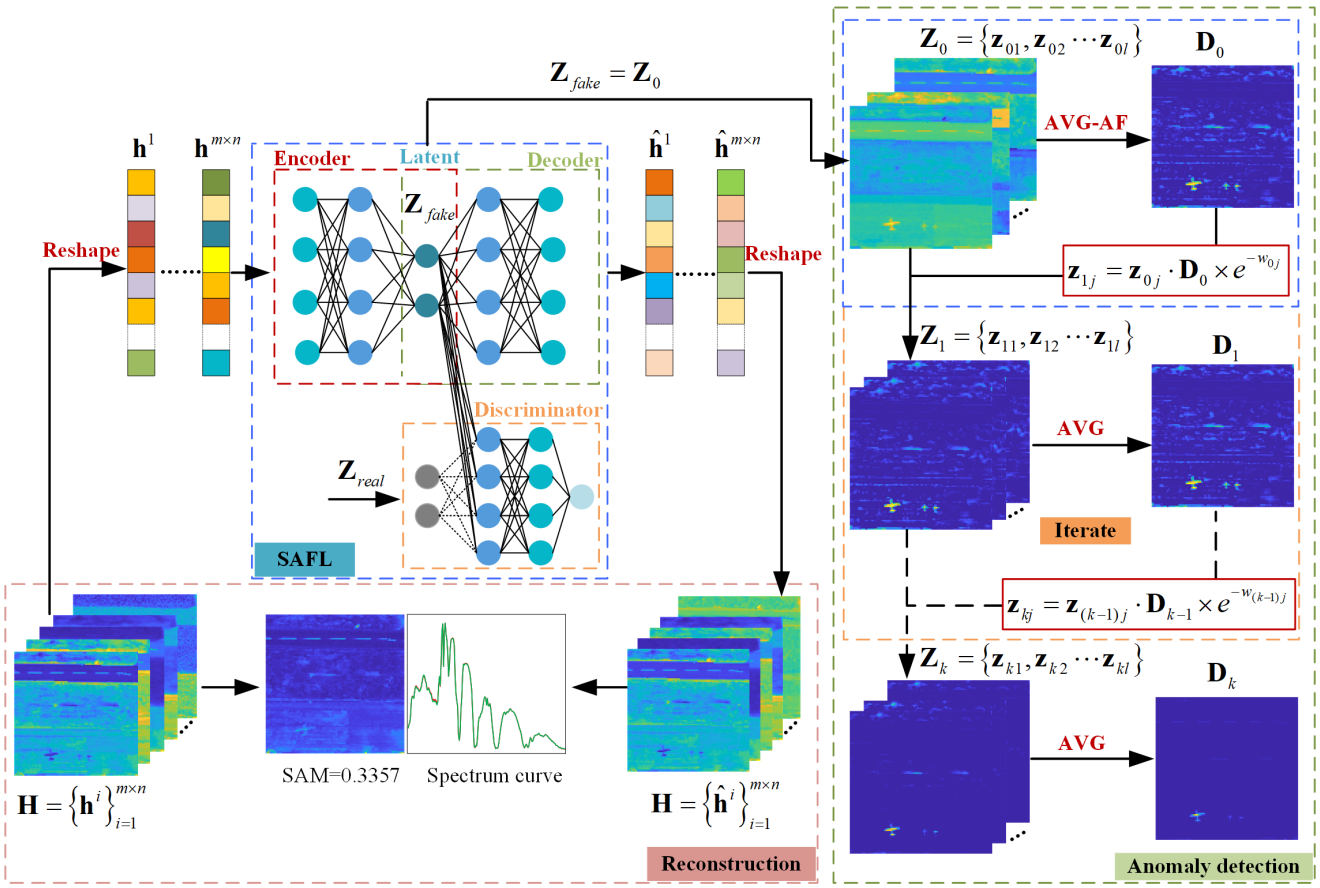


Fig. 1. Schematic of the proposed SAFL-based anomaly detection framework in HSIs.

the network architecture is designed with the consideration of the unique characteristic in HSIs.

III. Proposed Method

The schematic of the proposed hyperspectral anomaly detection method is shown in Fig. 1. It can be observed that the SAFL-based anomaly detection method consists of two mainly components, i.e., feature extraction and iterative optimization. Let $\mathbf{H} = \{\mathbf{h}^i\}_{i=1}^{M\times N}$ represent a normalized HSI, where $\mathbf{h}^i$ is the ith spectral vector with L spectral channels, i.e., $\mathbf{h}^i = [h_1^i, h_2^i, \ldots, h_L^i]$. Note that all values in $\mathbf{H} = \{\mathbf{h}^i\}_{i=1}^{M\times N}$ are in the range of $(-1, 1)$ after normalizing.

A. Spectral Adversarial Feature Learning

As shown in Fig. 1, the proposed SAFL model is an unsupervised neural network containing encoder, decoder, and discriminator. For a given hyperspectral data $\mathbf{H}$, the encoder portion of the network learns to express the input data $\mathbf{H}$ as a latent code $\mathbf{Z}_{\text{fake}}$. The decoder portion is responsible for reconstructing the input data $\mathbf{H}$ as $\hat{\mathbf{H}}$ according to the hidden code $\mathbf{Z}_{\text{fake}}$. Then, the discriminator fixed between the encoder and the decoder is trained to discriminatively distinguish whether its input data are generated by the encoder or sampled from the prior distribution specified by the user, after it gives the discriminant probability. The network architecture and the learning process are as follows.

1) Network Architecture: For the input data $\mathbf{H}$, the encoder maps it into a latent code $\mathbf{Z}_{\text{fake}}$ by using the trainable parameter set $\mathbf{W}_{hz}$, $\mathbf{b}_{hz}$, where $\mathbf{W}_{hz}$ denotes weight and $\mathbf{b}_{hz}$ is bias

$$\mathbf{Z}_{\text{fake}} = \sigma(\mathbf{H}\mathbf{W}_{hz} + \mathbf{b}_{hz}). \tag{1}$$

Here, $f(\cdot)$ indicates the activation function, which is generally the rectified linear units (ReLUs) [29] in AAE. ReLU is defined as $f(\mathbf{x}) = \max(0, \mathbf{x})$. As known in [30], the negative input is limited to zero, and the positive part is retained, which bounds the impact of negative input and leads to sparsity in neural networks. However, negative inputs may include useful information that benefits the training. As a variant of ReLU, the leaky ReLU (LReLU) adopts a linear function to compress the negative input [30] and retains the negative part of the information to a certain extent. The LReLU activation function can be defined as

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x < 0 \end{cases} \tag{2}$$

where x is the input of LReLU and α is a predefined constant parameter in the range of $(0, 1)$, which controls the slope of the negative part. Additionally, LReLU becomes ReLU when $\alpha = 0$. In this article, we choose $\alpha = 0.2$.

Since the distribution of input in each layer during training changes with the parameters of the previous layer, training

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in deep neural networks becomes complicated. This problem, sometimes referred to as internal covariate shift, leads to small changes in input parameters of each layer, which may amplify into more massive changes in the network. Batch normalization (BN) [31], which is proposed to address the internal covariate shift, has attracted great concern since it appeared. Regarded as a strategy to regularize the network without parameter adjustment, BN can be applicated in any layer in neural networks and avoid overfitting. Several recent studies in different fields [31], [32] have revealed its effectiveness in decreasing iteration epochs and internal covariate shift. Besides, limited availability of training HSIs in anomaly detection leading to overfitting during training is a common topic. Based on the abovementioned facts, BN is immediately employed before the LReLU to generate activation with a stable distribution and accelerate the training. Thus, LReLU activation function can be replaced as

$$f(x) = \begin{cases} \mathrm{BN}(x), & x > 0 \\ \alpha \mathrm{BN}(x), & x < 0 \end{cases} \tag{3}$$

where $\mathrm{BN}(x)$ denotes the BN layer applied in the input x and $x = \sigma(Wy + b)$. Here, x is the input of each fully connected layer. Moreover, the expression of $\mathrm{BN}(x)$ is as follows:

$$\mathrm{BN}(x) = \frac{x - E[x]}{\sqrt{\mathrm{var}[x] + \varepsilon}} \tag{4}$$

where $E[x]$ and $\mathrm{var}[x]$ denote the mean and variance of minibatch, respectively. Here, ε is defined as $1\mathrm{e}{-5}$ to avoid zero denominators in our architecture.

The decoder maps the hidden code $\mathbf{Z}_{\mathrm{fake}}$ back to the reconstruction $\hat{\mathbf{H}}$

$$\hat{\mathbf{H}} = \sigma(\mathbf{Z}_{\mathrm{fake}}\mathbf{W}_{zh} + \mathbf{b}_{zh}) \tag{5}$$

where σ is the activation function of the decoder and also LReLU here. $\mathbf{W}_{zh}$ and $\mathbf{b}_{zh}$ denote the weight and bias of the decoder, respectively.

The hidden codes obtained from the encoder and the prior distribution given by the user are fed to discriminator simultaneously. The prior distribution is chosen as the normal distribution in our model due to the hypothesis that HSIs obey the Gaussian probability in [11]. In the discriminator, except that the activation function of the last layer is sigmoid $f(\mathbf{x}) = (1 + \exp(-\mathbf{x}))^{-1}$, the activation functions of all the other layers are LReLU.

2) Loss: The discriminator and encoder make up a GAN in the training phase of which the encoder changes into a generator to ensure that the generated hidden code can fool the discriminator. Also, the discriminator D aims to tell part its input coming from the encoder or the prior normal distribution. Hence, the generator G (encoder) tries to minimize this objective against an adversary discriminator that tries to maximize it

$$L_{\mathrm{GD}} = \min_G \max_D E[\log D(\mathbf{Z}_{\mathrm{rea\,l}})] + E[\log(1 - E(\mathbf{H}))]. \tag{6}$$

The proposed SAFL model learns to minimize the loss function concerning the trainable parameters $\{\mathbf{W}_{hz}, \mathbf{W}_{zh}, \mathbf{b}_{hz}, \mathbf{b}_{zh}\}$. Generally, it uses a learning algorithm, such as stochastic gradient descent (SGD) [33] with backpropagation to optimize parameters. The Euclidean ($l2$) norm is selected as the cost function to update the encoder and decoder in AE or AAE. Unfortunately, the reconstruction loss is typically a pixelwise cross entropy between the input and output images, which often leads to the generated images to be blurry and wrecks the inputs. Therefore, by taking the spectral consistency between the input and the reconstructed spectral vectors into consideration, we introduce an additional constraint into the reconstruction loss. Spectral angle mapper (SAM) [34] is an index to evaluate the spectral similarity between the two input spectral vectors. Moreover, the lower the value of SAM, the more similar the two spectral vectors. The definition of SAM can be described as

$$\mathrm{SAM} = \frac{1}{mn}\left(1 - \frac{1}{\pi}\arccos\left(\frac{\mathbf{H}\hat{\mathbf{H}}}{\|\mathbf{H}\|_2\|\hat{\mathbf{H}}\|_2}\right)\right). \tag{7}$$

Here, $\mathbf{H} \in \mathbb{R}^{m\times n\times L}$ and $\hat{\mathbf{H}} \in \mathbb{R}^{m\times n\times L}$ are the two input samples. Based on the abovementioned analysis, we optimize the encoder and decoder jointly by using (8) so that the spatial and spectral feature information is considered in the process of feature extraction

$$L_{ED'} = \|\mathbf{H} - \hat{\mathbf{H}}\|_2 + \lambda\frac{1}{mn}\left(1 - \frac{1}{\pi}\arccos\left(\frac{\mathbf{H}\hat{\mathbf{H}}}{\|\mathbf{H}\|_2\|\hat{\mathbf{H}}\|_2}\right)\right). \tag{8}$$

3) Learning Process: The SAFL model is trained in two phases during each minibatch: First, the encoder and decoder update their parameters $\{\mathbf{W}_{hz}, \mathbf{W}_{zh}, \mathbf{b}_{hz}, \mathbf{b}_{zh}\}$ to minimize the spatial and spectral reconstruction error. Subsequently, the discriminator and the encoder (which is also the generator) make up a GAN, of which the discriminator generates the discriminative probability of whether its input is true or false. Also, according to the discriminative probability, the parameters of the generator are updated to fool the discriminator. In the next minibatch, the training process will continue on the basis of the parameters obtained from the previous minibatch, that is, the update of parameters $\{\mathbf{W}_{hz}, \mathbf{W}_{zh}, \mathbf{b}_{hz}, \mathbf{b}_{zh}\}$ is determined by alternate training of encoder, decoder, and discriminator in each minibatch. By training the encoder, decoder, and discriminator jointly for thousands of epochs and when the discriminator cannot distinguish whether its input is true or false, the parameters are optimized and a model for feature extraction is obtained. As mentioned earlier, we are interested in the output of the encoder because the hidden code is considered as the latent representation of the input high-dimensional spectral vectors. Twenty features of the San Diego scene extracted by SAFL is shown in Fig. 2.

B. Iterative Optimization

Based on the observation that anomalies in HSIs usually occupy fewer pixels compared with background and the relevance among anomalous pixels is higher, we introduce a morphological AF proposed in [24] to enhance anomalies. Inspired by our previous work [35], the anomaly enhancement is achieved in two phases: first, fuse all the extracted features into one by the AVG method, and then, extract small areas and remove the interference background information by calculating

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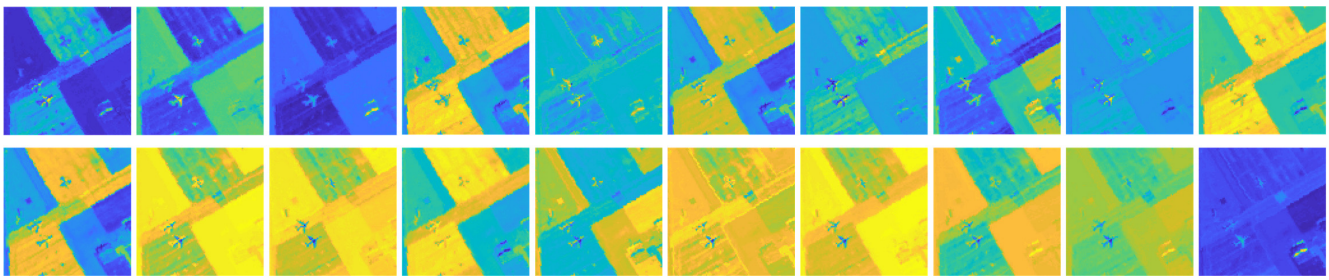


Fig. 2. Intermediate results of hidden codes in visual by the SAFL model with 20 hidden nodes.

Algorithm 1 Iterative Optimization to Suppress Noise

Input: $\mathbf{D}_0$: the initial detection map; O: the threshold; $\mathbf{Z}_0 = \{\mathbf{z}_{01}, \mathbf{z}_{02} \cdots \mathbf{z}_{0\,l}\}$: spectral features extracted by SAFL.

Output: the final detection map.

1: **while** (1) **do**

2: Calculate the weight vector $\mathbf{W}_{(k-1)j} = \{w_{(k-1)j}, w_{(k-1)j} \cdots w_{(k-1)j}\}$ corresponding to features in the $(k-1)$th iteration.

3: Update features $\mathbf{Z}_k = \{\mathbf{z}_{k1}, \mathbf{z}_{k2} \cdots \mathbf{z}_{kl}\}$ of the kth iteration by using weight vector and the $(k-1)$th detection map:

$$\mathbf{z}_{kj} = \mathbf{z}_{(k-1)j} \cdot \mathbf{D}_{k-1} \times e^{-w_{(k-1)j}} \tag{9}$$

4: Calculate the detection map of the kth iteration by applying averaging fusion of features:

$$\mathbf{D}_k = \frac{1}{l} \sum_{j=1}^{l} \mathbf{z}_{kj} \tag{10}$$

5: **if** SSIM $(\mathbf{D}_{k+1}, \mathbf{D}_k) > O$ **then**

6: stop;

7: **else**

8: continue;

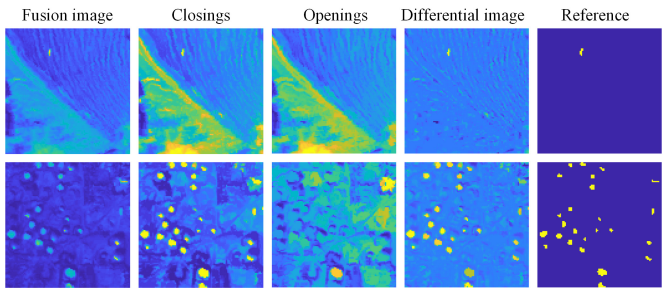


Fig. 3. Steps of obtaining an anomaly enhancement result in visual.

the attribute openings and closings of the fused feature. Two visual results are shown in Fig. 3.

The anomaly enhancement map can preserve anomalies, but the false alarm rate is too high to apply to engineer practice. Furthermore, it can be observed from the differential map in Fig. 3 that the noise, which is also the background of mistaken anomalies, accounts for a considerable proportion. To suppress noise, we investigate an adaptive weighting method to suppress noise in the differential map through iterative optimization. In particular, the adaptive weighting method can be described as two steps, i.e., compute ST matrix of each feature and calculate the contribution of each feature.

Usually, edges and corners are essential features of images and play a vital role in understanding and analyzing images. ST [36], which can be employed to describe the structural information of an image, has been exploited in the fields of image processing [37], [38]. For the strong representational capability of ST, we apply it to compute the contribution rate corresponding to each feature. ST is the structural matrix of the given image z_i, where $\mathbf{z}_i^x$ and $\mathbf{z}_i^y$ are horizontal and vertical gradients, respectively. The matrix can be described as follows:

$$T = \begin{bmatrix} \left(\mathbf{z}_i^x\right)^2 & \mathbf{z}_i^{xy} \\ \mathbf{z}_i^{yx} & \left(\mathbf{z}_i^y\right)^2 \end{bmatrix}. \tag{11}$$

Due to the fact that the matrix is semidefinite, it can be decomposed into the eigenvalues λ_1 and λ_2 and corresponding eigenvectors v_1 and v_2 as shown in following equation:

$$T = [v_1, v_2] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}. \tag{12}$$

Here, λ_1 and λ_2 can describe the structure of the given image. If $\lambda_1 \approx \lambda_2 \approx 0$, it belongs to smooth areas with tiny variations. For edge areas with large variations concentrated in a significant direction, λ_1 and λ_2 are satisfied with $\lambda_1 \gg \lambda_2 \approx 0$ or $\lambda_2 \gg \lambda_1 \approx 0$. Corner areas mean that both λ_1 and λ_2 are larger, i.e., $\lambda_2 \gg \lambda_1 \gg 0$.

Assume that λ_1^i, λ_2^i are the eigenvalues of the ith differential map and v_1^j and v_2^j are the corresponding eigenvector of the ith differential feature. As stated earlier, the eigenvalue of ST can represent the structural information of a given image effectively. In addition, the larger the eigenvalue, the more structural information the feature owns. Therefore, the weight of the ith feature can be calculated by

$$\omega_i = \frac{\lambda_1^i}{\sum_{i=1}^{l} \lambda_1^i} \tag{13}$$

where l is the total number of features obtained by SAFL. As shown in Fig. 2, features with more background information own larger weight because anomalies are sparse and have a lower probability of occurring. Based on the observed fact, we constrain weight with a negative exponential function of which decreasing character is between 0 and 1. Therefore,

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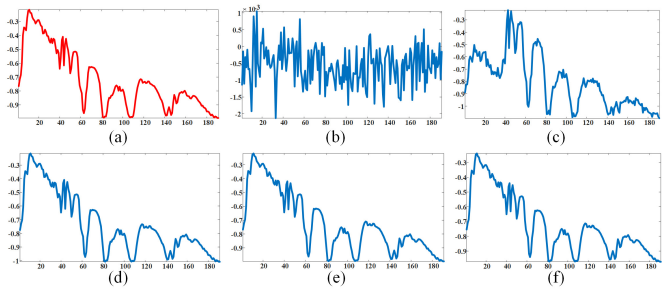


Fig. 4. Reconstructions of the input pixel in different iteration epochs. (a) Input spectrum. (b)–(f) Reconstructions of (a) in epochs 1, 100, 1000, 5000, and 10000, respectively. The vertical axis indicates the normalized reflectance, whereas the horizontal axis implies band number.

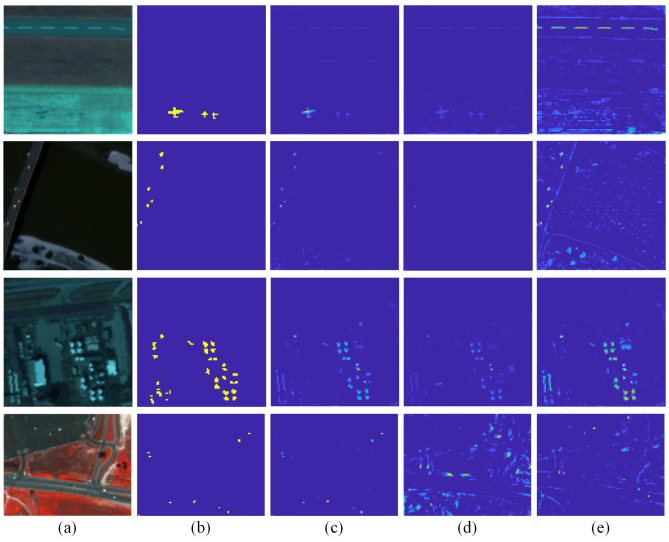


Fig. 5. Detection maps of the compared feature extraction methods. (a) Color composites of four scenes. (b) Reference map. (c) SAFL. (d) AE. (e) AAE.

it is equivalent to that the feature with anomalies has larger weight, while the more background in feature has smaller weight. The details of the iterative optimization algorithm are shown in Algorithm 1. Note that structural similarity is able to evaluate the similarity of two images, $\mathbf{D}_x$ and $\mathbf{D}_y$. Therefore, we propose a stopping rule based on the structural similarity index (SSIM) [39] to adaptively stop iteration. The value of SSIM can be computed as

$$\mathrm{SSIM}(\mathbf{D}_x, \mathbf{D}_y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \qquad (14)$$

where μ_x and μ_y are the mean values of $\mathbf{D}_x$ and $\mathbf{D}_y$, σ_x^2, and σ_y^2 are the variance. σ_{xy} is the covariance between $\mathbf{D}_x$ and $\mathbf{D}_y$. c_1 and c_2 are constants. The similarity score computed by SSIM is between 0 and 1, and a higher score means a higher similarity. Therefore, in the noise suppression algorithm, if the previous detection map is very similar to the current one, it can be explained that the current detection map is excellent and the loop can be terminated. Here, we stop the noise suppression algorithm when the similarity between the latest two detection maps is more than the predefined threshold O.

TABLE I

INFLUENCE OF DIFFERENT NUMBERS OF UNITS IN THE THIRD LAYER ON THE DETECTION PERFORMANCE FOR EACH HSI

Number of Units	10	20	30	40	50
ABU-a1	0.93506	**0.98054**	0.97156	0.98128	0.97322
ABU-a2	**0.99067**	0.98049	0.98671	0.98275	0.98149
ABU-a3	0.93121	**0.97131**	0.95141	0.96879	0.96982
ABU-a4	0.95825	**0.99708**	0.98133	0.99601	0.99070
Average	0.95380	**0.98235**	0.97275	0.98221	0.97881
Number of Units	10	20	30	40	50
ABU-b1	0.96889	**0.98930**	0.98150	0.98769	0.98134
ABU-b2	0.94612	**0.99518**	0.97962	0.98687	0.99011
ABU-b3	0.98748	0.99898	0.98654	0.98037	**0.99954**
ABU-b4	0.97271	0.98739	**0.99242**	0.98789	0.98340
Average	0.96880	**0.99271**	0.98502	0.98570	0.98859
Number of Units	10	20	30	40	50
ABU-u1	0.99325	**0.99734**	0.99287	0.98655	0.92528
ABU-u2	0.96796	0.99619	**0.99819**	0.99322	0.99341
ABU-u3	0.96687	**0.99762**	0.99752	0.99162	0.99395
ABU-u4	0.98972	**0.99675**	0.99565	0.99560	0.98575
ABU-u5	0.96775	0.98914	0.98157	0.98684	**0.98923**
Average	0.97711	**0.99541**	0.99316	0.99077	0.97752
Number of Units	10	20	30	40	50
SanDiego	0.98838	**0.99706**	0.97978	0.98068	0.96176
Number of Units	10	20	30	40	50
HYDICE	0.98854	**0.99996**	0.99470	0.99989	0.99962

IV. EXPERIMENTAL EVALUATION

A. Data Sets' Description

In this section, our proposed SAFL-based anomaly detection is evaluated on three real hyperspectral data sets, including 15 HSIs captured over different scenes.

1) Airport-Beach-Urban (ABU)[1]*:* The ABU data set contains airport, beach, and urban scenes, and most of them were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. Only one of the beach scenes is captured by the Reflective Optics System Imaging Spectrometer (ROSIS) Sensor. Thirteen HSIs in this data set contain different spatial pixels and different spectral dimensions. The detailed description is in [40].

2) HYDICE Urban[2]*:* These data are a standard and widely used HSI to evaluate the performance of anomaly detection, which is captured from an urban by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. In the experiment, a subimage containing point anomalies of size 80 × 100 with 162 spectral bands is used.

[1] http://xudongkang.weebly.com/

[2] http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetsArticleView/tabid/9254/Article/476681/hypercube

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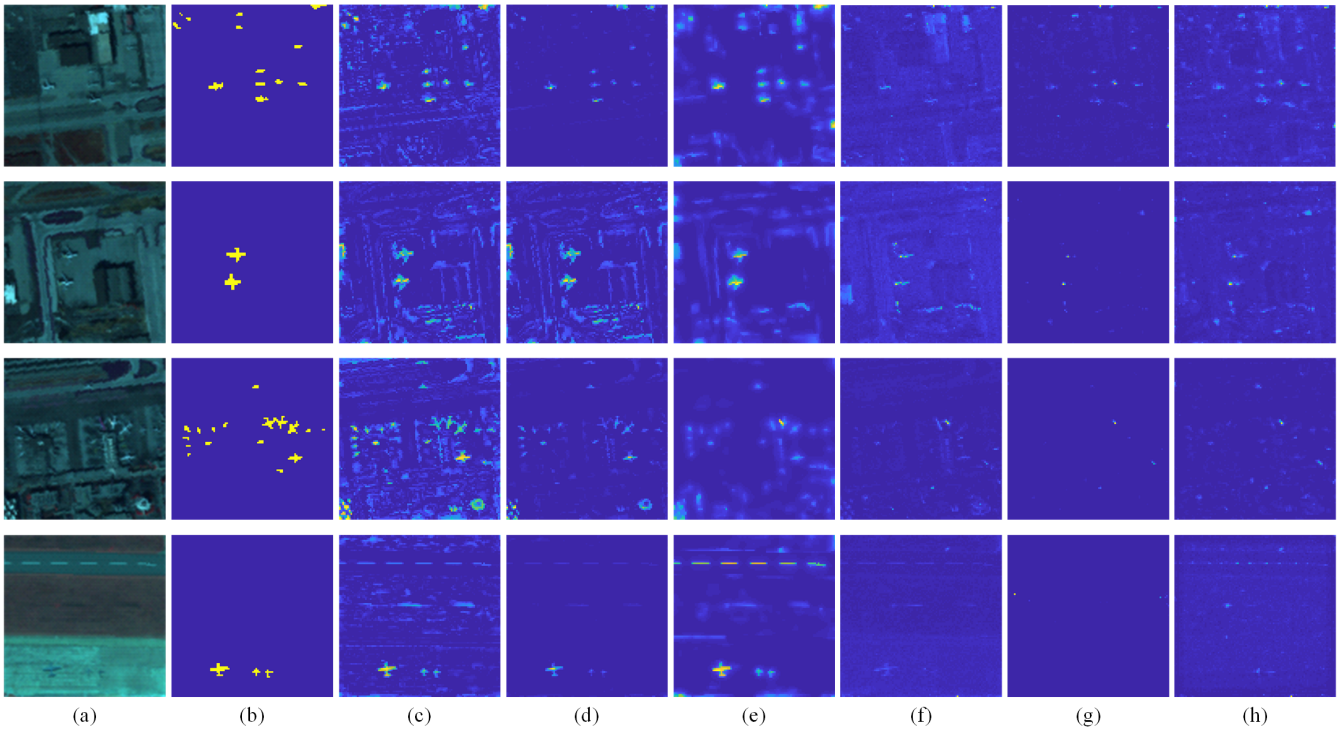


Fig. 6. Detection maps of the compared methods. (a) Color composites of airport scene in ABU data set. (b) Reference map. (c) SAFL-ANE. (d) SAFL-NSS. (e) AED. (f) RX. (g) LRX. (h) CRD.

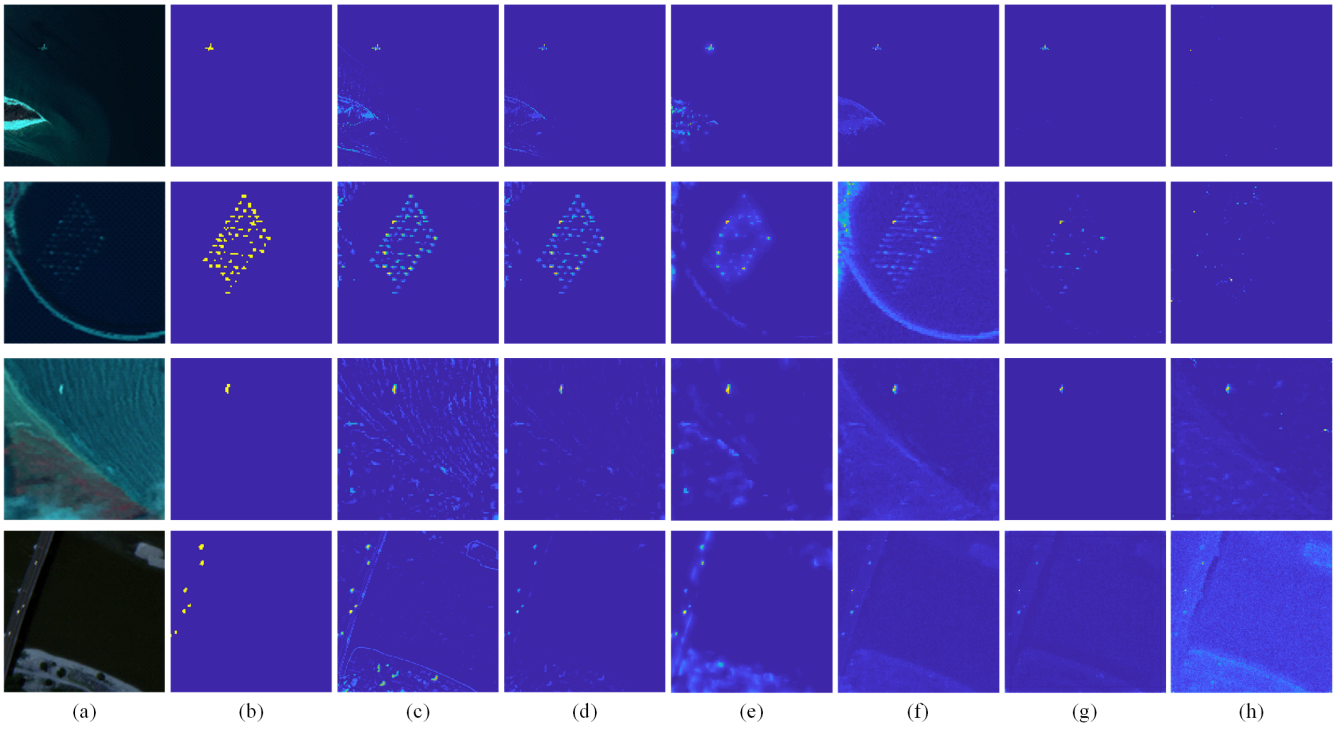


Fig. 7. Detection maps of the compared methods. (a) Color composites of beach scene in ABU data set. (b) Reference map. (c) SAFL-ANE. (d) SAFL-NSS. (e) AED. (f) RX. (g) LRX. (h) CRD.

3) San Diego: This data set is collected by the AVIRIS sensor and it covers the area of the San Diego airport, CA, USA. This real HSI has an image size of 100 $\times$ 100 $\times$ 189, in which the noisy bands have been removed. There are three aircrafts in different directions that are regarded as interesting objects in the scene. This data set is frequently applied in the anomaly detection problem.

B. Evaluation and Experimental Setup

1) Evaluation: In the detection field, the commonly used metric to evaluate the performance is the receiver operating

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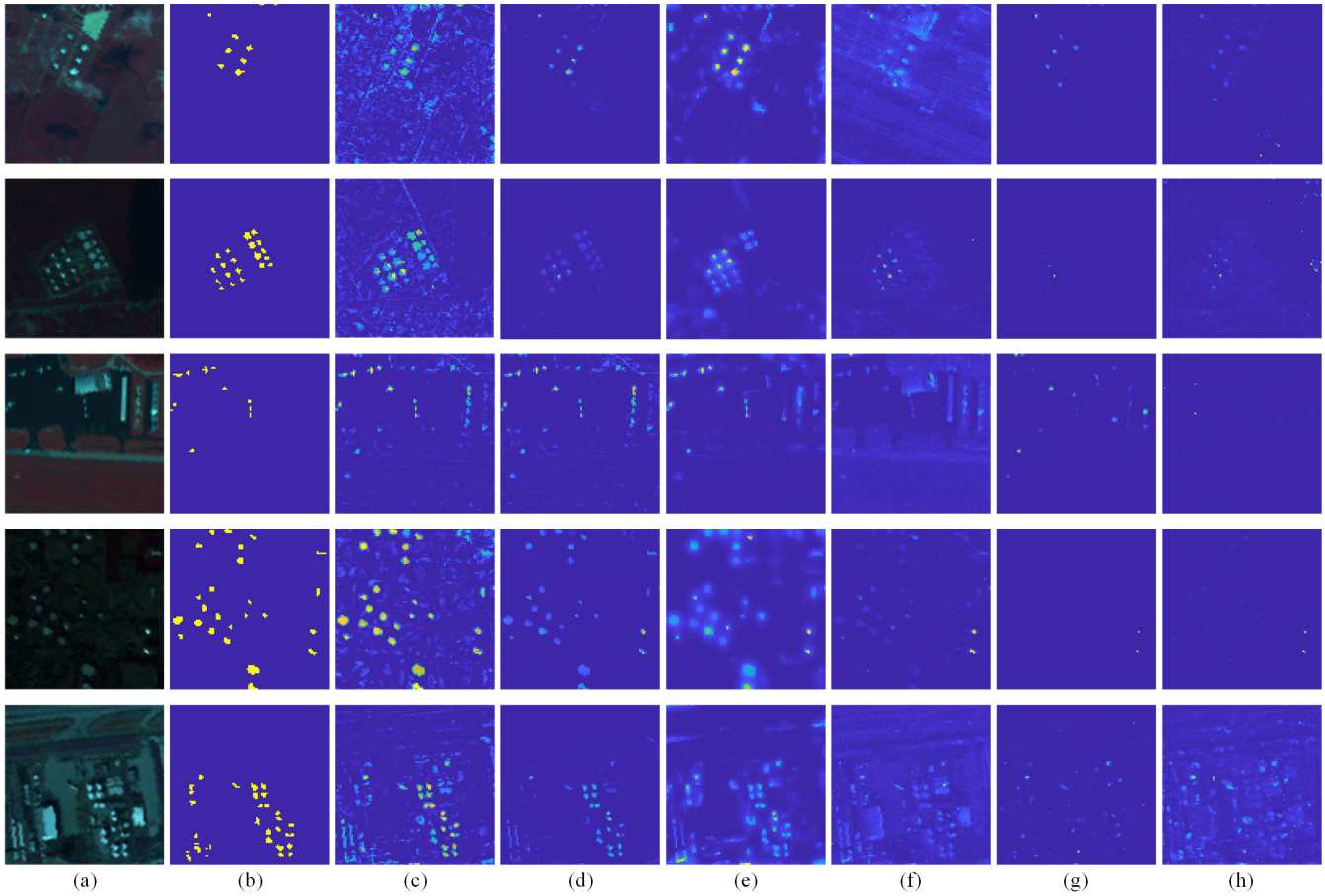


Fig. 8. Detection maps of the compared methods. (a) Color composites of urban scene in ABU data set. (b) Reference map. (c) SAFL-ANE. (d) SAFL-NSS. (e) AED. (f) RX. (g) LRX. (h) CRD.

characteristic (ROC) curve and the area under the ROC curve (AUC) [41]. The ROC curve describes the detection performance with the true positive rate (P_d) and the false positive rate (P_f) at each threshold (τ). These three parameters, P_d, P_f, and τ, can be applied to generate three types of ROC curves. Among them, the ROC curve of (P_d and P_f) describes the tradeoff between P_d and P_f, and the closer the curve is to the upper left corner, the higher the AUC value. The ROC curve of (P_d and τ reports the detection capability at each threshold τ, while the ROC of (P_f and τ indicates the false alarm rate of the anomaly detection, that is, the larger the value of the area under the curve of (P_d and P_f) and (P_d and τ, the better the detection performance. The smaller the area under the curve of (P_f and τ, the lower the probability of false detection. Besides, the SAM index is also used to estimate the reconstruction capability of our proposed method.

2) Experimental Setup : We set the depth of the proposed encoder, decoder, and discriminator is 3, and the first two layers are two fully connected layers with 500 units, and the number of the units in the third layer is set as 20. Table I reports the influence of different numbers of units in the third layer for the detection performance. It is observed from Table I that the proposed method achieves better detection performance when the number of hidden units in the third layer is 20 for almost all test data sets. In other words, the features extracted by SAFL are restricted to 20, which means that the original hyperspectral data with hundreds of bands are compressed to 20 dimensions. The batch size is set as the number of spatial pixels in each HSI and training is stopped after 10 000 epochs. The learning rate is 0.001 for the encoder and decoder and 0.0001 for the discriminator. Furthermore, all the parameters are optimized by SGD with backpropagation. Parameters introduced earlier are default in our experiments, and they can be tuned by the user for the optimal results.

C. Component Analysis

1) Effectiveness Evaluation of Reconstruction: In the proposed SAFL model, the better the performance of the reconstruction, the more information the features in the deep latent space embodies. The quality of the deep spectral features is evaluated by the reconstruction error between the input spectral vector and the reconstructed one. We randomly choose one pixel in one of the selected data sets and draw the spectral curve of this pixel during the training. As shown in Fig. 4, the SAFL model enables to progressively reconstruct the original spectral curve after 5000 epochs. Thus, the latent code can be treated as an effective feature for the input. Besides, it also extraordinarily proves the capability to extract features of our proposed SAFL model.

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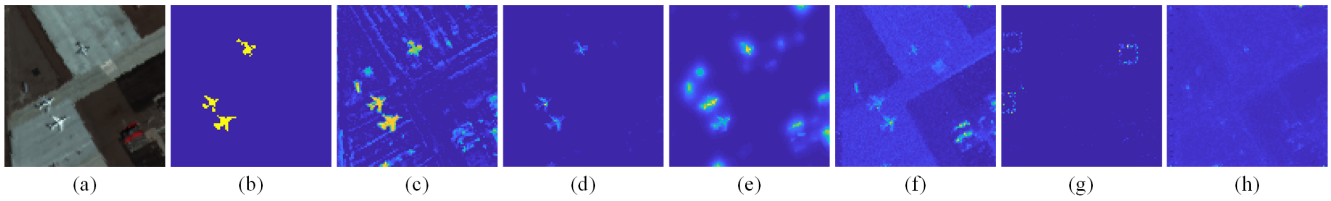


Fig. 9. Detection maps of the compared methods. (a) Color composites of San Diego data set. (b) Reference map. (c) SAFL-ANE. (d) SAFL-NSS. (e) AED. (f) RX. (g) LRX, (h) CRD.

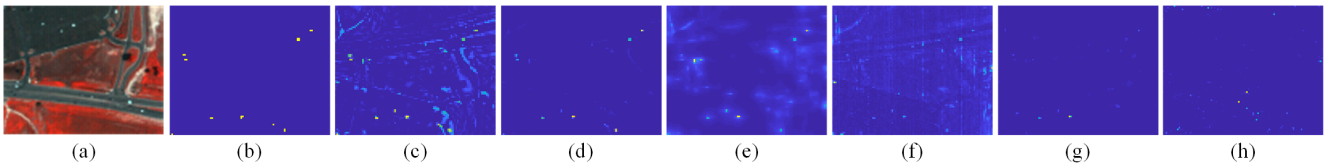


Fig. 10. Detection maps of the compared methods. (a) Color composites of HYDICE urban data set. (b) Reference map. (c) SAFL-ANE. (d) SAFL-NSS. (e) AED. (f) RX. (g) LRX. (h) CRD.

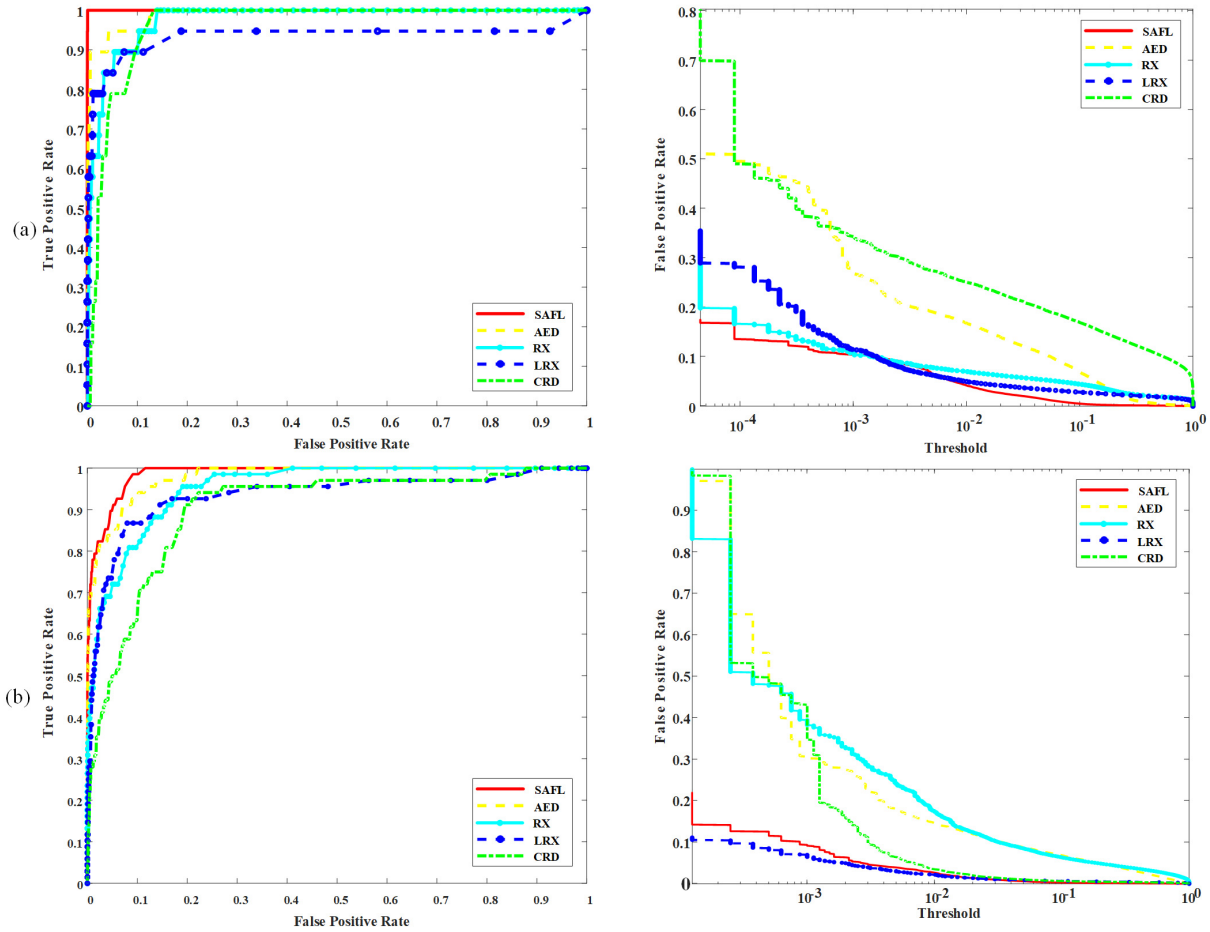


Fig. 11. ROC curves of (P_d and P_f) and (P_f and τ) obtained by the compared methods on (a) fourth beach scene in ABU data set and (b) HYDICE urban data set.

2) Effectiveness Evaluation of SAFL: To illustrate the superior performance of the proposed SAFL method compared with AE and AAE, we replace SAFL with AE and AAE in our proposed anomaly detection framework. By analyzing the anomaly detection results in different data sets using SAFL, AE, and AAE, we can validate the effectiveness of the specially designed architecture in SAFL. As reported in Table II, we calculate the AUC scores of (P_d, P_f), and (P_d, τ) based on SAFL, AE, and AAE. In terms of the detection accuracy, SAFL is capable of achieving the best

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TABLE II
EVALUATION AUC SCORES AND FALSE ALARM RATE OF THE SAFL, AE, AND AAE-BASED DETECTION METHODS

Images	(P_d, P_f)			(P_f, τ)		
	SAFL	AE	AAE	SAFL	AE	AAE
ABU-a1	**0.98054**	0.85697	0.87469	**0.00470**	0.01312	0.03451
ABU-a2	**0.98049**	0.90749	0.96977	0.03589	**0.01430**	0.01609
ABU-a3	**0.97131**	0.91159	0.92195	0.00996	**0.00289**	0.01361
ABU-a4	**0.99708**	0.99610	0.94410	**0.00212**	0.00530	0.02765
Average	**0.98235**	0.91804	0.92763	0.01317	**0.00890**	0.02296
Images	(P_d, P_f)			(P_f, τ)		
	SAFL	AE	AAE	SAFL	AE	AAE
ABU-b1	**0.98930**	0.95709	0.95365	0.00226	**0.00093**	0.00296
ABU-b2	**0.99518**	0.99217	0.99083	0.00381	0.00547	**0.00121**
ABU-b3	0.99898	0.35639	**0.99941**	**0.00447**	0.01940	0.00589
ABU-b4	**0.98739**	0.94358	0.97961	0.00224	**0.00026**	0.01941
Average	**0.99271**	0.81231	0.98087	**0.00320**	0.00652	0.00737
Images	(P_d, P_f)			(P_f, τ)		
	SAFL	AE	AAE	SAFL	AE	AAE
ABU-u1	**0.99734**	0.99720	0.98922	**0.00299**	0.00517	0.00941
ABU-u2	0.99619	**0.99939**	0.96806	0.00327	**0.00022**	0.04229
ABU-u3	**0.99762**	0.98613	0.99211	0.01183	**0.00352**	0.00742
ABU-u4	**0.99675**	0.97729	0.99489	**0.00317**	0.06108	0.02030
ABU-u5	**0.98914**	0.97157	0.98026	**0.00337**	0.00389	0.00915
Average	**0.99541**	0.98631	0.98491	**0.00493**	0.01400	0.01771
Images	(P_d, P_f)			(P_f, τ)		
	SAFL	AE	AAE	SAFL	AE	AAE
San Diego	**0.99706**	0.94483	0.95946	**0.00207**	0.02144	0.02841
Images	(P_d, P_f)			(P_f, τ)		
	SAFL	AE	AAE	SAFL	AE	AAE
HYDICE	**0.99996**	0.82362	0.92185	**0.00126**	0.02684	0.00962

TABLE III
EVALUATION AUC SCORES AND FALSE ALARM RATE OF THE MAIN COMPONENTS OF OUR METHOD

Images	(P_d, P_f)		(P_f, τ)	
	ANE	NSS	ANE	NSS
ABU-a1	0.98370	0.98054	0.03351	0.00470
ABU-a2	0.99013	0.98049	0.04986	0.03589
ABU-a3	0.97044	0.97131	0.05007	0.00996
ABU-a4	0.99813	0.99708	0.02187	0.00212
Average	0.98560	0.98235	0.03883	0.01317
Images	(P_d, P_f)		(P_f, τ)	
	ANE	NSS	ANE	NSS
ABU-b1	0.98940	0.98930	0.00578	0.00226
ABU-b2	0.99532	0.99518	0.00516	0.00381
ABU-b3	0.99950	0.99898	0.01568	0.00447
ABU-b4	0.98937	0.98739	0.01274	0.00224
Average	0.99340	0.99271	0.00984	0.00320
Images	(P_d, P_f)		(P_f, τ)	
	ANE	NSS	ANE	NSS
ABU-u1	0.99242	0.99734	0.04798	0.00299
ABU-u2	0.99316	0.99619	0.02891	0.00327
ABU-u3	0.99832	0.99762	0.01327	0.01183
ABU-u4	0.99651	0.99675	0.03198	0.00317
ABU-u5	0.98860	0.98914	0.02412	0.00337
Average	0.99380	0.99541	0.02925	0.00493
Images	(P_d, P_f)		(P_f, τ)	
	ANE	NSS	ANE	NSS
San Diego	0.99574	0.99706	0.03981	0.00207
Images	(P_d, P_f)		(P_f, τ)	
	ANE	NSS	ANE	NSS
HYDICE	0.99965	0.99996	0.02212	0.00126

performance on different real hyperspectral data sets. As far as the false alarm rate, our proposed SAFL model can obtain lower values on the premise of ensuring detection accuracy. Note that the data sets include points anomalies and collective anomalies. The detection maps of SAFL, AE, and AAE are shown in Fig. 5. It can be observed that the proposed SAFL model can achieve more meaningful features, and the performance is outstanding in contrast with AE and AAE. The best scores in Table II are highlighted in bold for each data set.

3) Effectiveness Evaluation of Iterative Optimization: To further validate the influence of the proposed iterative optimization, we provide the objective results and subjective analysis of our proposed anomaly detection system step by step. The main components in the iterative optimization are anomaly enhancement (ANE) and noise suppression (NSS), so we analyze the effects of these two components by the AUC scores of (P_d, P_f), and (P_f, τ). The results of each component for different data sets are shown in Table III. Note that ANE denotes the anomaly enhancement step and NSS implies the noise suppression step. It is obvious that the anomaly enhancement step can detect almost all anomalies but with a larger false alarm rate, while after the noise suppression step, the false alarm rate reduces significantly because of the process of continuous iterative optimization. Meanwhile, the detection rate has slight change. Fortunately, most detection rates after the noise suppression is increased, and the extent of the decline is tiny. In other words, these two components have a positive impact on the performance of anomaly detection. Columns 3 and 4 in Figs. 6–10 show the detection map obtained by each component in the anomaly detection system. Table III reports the AUC scores of (P_d, P_f), and (P_f, τ) corresponding to each component, and they are consistent with Figs. 6–10.

D. Detection Results

In this section, the anomaly detection performance of the proposed SAFL method is evaluated and compared with three typical anomaly detectors: RX, LRX, CRD, and one of the state-of-the-art detector AED. As it is known, the window size in the LRX and CRD is significant, and different window sizes produce different detection results. Thus, in our experiments, the optimal window size of LRX and CRD for each data set is selected for the optimal detection performance. For the

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TABLE IV
EVALUATION AUC SCORES OBTAINED BY DIFFERENT METHODS ON THE ABU, SAN DIEGO, AND HYDICE DATA SETS

Images	Proposed	AED	RX	LRX	CRD
ABU-a1	0.98054	**0.99226**	0.82259	0.89402	0.96071
ABU-a2	0.98049	**0.99360**	0.84059	0.85155	0.99360
ABU-a3	0.97131	**0.97562**	0.92899	0.92576	0.94722
ABU-a4	**0.99708**	0.99534	0.95206	0.88793	0.82168
Average	0.98235	**0.98920**	0.88606	0.88981	0.93080
Images	Proposed	AED	RX	LRX	CRD
ABU-b1	0.98930	**0.99740**	0.98281	0.97495	0.96720
ABU-b2	**0.99518**	0.95495	0.91060	0.96058	0.92214
ABU-b3	0.99898	0.99968	0.99985	**0.99997**	0.99914
ABU-b4	**0.98739**	0.97935	0.95335	0.93479	0.90093
Average	**0.99271**	0.98285	0.96165	0.96757	0.94735
Images	Proposed	AED	RX	LRX	CRD
ABU-u1	0.99734	**0.99815**	0.99064	0.99279	0.99279
ABU-u2	**0.99619**	0.78570	0.99463	0.62947	0.93042
ABU-u3	**0.99762**	0.99754	0.95119	0.89849	0.93970
ABU-u4	**0.99675**	0.99116	0.98870	0.66332	0.95156
ABU-u5	**0.98914**	0.98452	0.96932	0.92745	0.93045
Average	**0.99541**	0.95141	0.97890	0.82231	0.94898
Images	Proposed	AED	RX	LRX	CRD
San Diego	**0.99706**	0.97966	0.95151	0.50815	0.83742
Images	Proposed	AED	RX	LRX	CRD
HYDICE	**0.99996**	0.98988	0.97637	0.93407	0.96206

TABLE V
EVALUATION FALSE ALARM RATES OBTAINED BY DIFFERENT METHODS ON THE ABU, SAN DIEGO, AND HYDICE DATA SETS

Images	Proposed	AED	RX	LRX	CRD
ABU-a1	**0.00470**	0.03293	0.04232	0.01336	0.03362
ABU-a2	0.03589	0.03011	0.05154	**0.00370**	0.02964
ABU-a3	0.00996	0.02249	0.01444	**0.00094**	0.01187
ABU-a4	0.00212	0.02425	0.02471	0.00584	**0.00079**
Average	0.01317	0.02744	0.03325	**0.00596**	0.01898
Images	Proposed	AED	RX	LRX	CRD
ABU-b1	0.00226	0.00410	0.00652	0.00155	**0.00076**
ABU-b2	0.00381	0.01227	0.04875	0.00551	**0.00282**
ABU-b3	0.00447	0.01161	0.02592	**0.00116**	0.01935
ABU-b4	**0.00224**	0.01998	0.02332	0.01939	0.11240
Average	**0.00320**	0.01199	0.02613	0.00690	0.03383
Images	Proposed	AED	RX	LRX	CRD
ABU-u1	**0.00299**	0.02292	0.05552	0.00389	0.00889
ABU-u2	0.00327	0.01437	0.01351	**0.00051**	0.01449
ABU-u3	0.01183	0.01379	0.03503	0.00515	**0.00229**
ABU-u4	0.00317	0.03069	0.01141	**0.00050**	0.00121
ABU-u5	**0.00337**	0.03058	0.04370	0.00918	0.04447
Average	0.00493	0.02247	0.03183	**0.00385**	0.01427
Images	Proposed	AED	RX	LRX	CRD
San Diego	**0.00207**	0.03441	0.07037	0.00501	0.05874
Images	Proposed	AED	RX	LRX	CRD
HYDICE	**0.00126**	0.03089	0.03798	0.00398	0.00474

AED method, we obtain the detection result of each data set according to the parameters and code given in [40]. Figs. 6–10 show the detection maps of different compared methods for each data set. To quantitatively evaluate the performance of the proposed method, the AUC scores of each compared method are presented in Table IV. The false alarm rate corresponding to each data set is shown in Table V. Please note that the best scores in Tables IV and V are highlighted in bold for each data set. Detection accuracy is the primary evaluation index to illustrate the performance of anomaly detection. With similar detection accuracy, we then choose the false alarm rate to evaluate different methods, and the smaller the false alarm rate, the better the method.

For the airport scene in the ABU data set, the average AUC score obtained by our proposed SAFL method is 0.98235, which is much higher than those three typical methods. However, it is 0.00685 smaller than the best approach, AED, which is 0.98920. However, the average false alarm rate of the proposed method is 0.01317 that is better than AED. Although the LRX method has the best false alarm rate, it also has a poor average AUC score. Our proposed method has a tradeoff performance in consideration of the average AUC score and the false alarm rate. Besides, the average false alarm rate of the proposed method is almost half as small as 0.02744 obtained by the AED method, although they have a similar detection accuracy.

For the beach scene in the ABU data set, the detection maps are shown in Fig. 7. It can be seen that the average AUC score, 0.99271, is largest among all the methods and the average false alarm rate 0.00320 is smallest compared with other algorithms. Viewed from the detection maps, it is not difficult to find that they are close to the reference map for each scene.

The detection results of the urban scene in the ABU data set are shown in Fig. 8. Compared with AED, RX, LRX, and CRD, our proposed method owns the detection accuracy 0.99541, which means that it can detect more anomalies with a lower false alarm rate that is crucial to evaluate the performance of anomaly detection. Note that the average false alarm rate 0.00385 obtained by LRX is lower than our proposed method. However, the detection accuracy of LRX is the worst, which means that it is not enabled to detection more anomalies.

For the San Diego (see Fig. 9) and HYDICE (see Fig. 10) data sets, the AUC scores and the false alarm rate obtained

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TABLE VI
RUNNING TIME FOR THE COMPARED METHODS ON DIFFERENT DATA SETS

Images	Proposed	AED	RX	LRX	CRD
ABU-a1	41.5000	0.1157	0.2189	62.7359	73.5380
ABU-a2	43.9844	0.1194	0.2008	54.7698	74.4010
ABU-a3	55.2969	0.1109	0.1895	54.7400	74.2396
ABU-a4	40.6094	0.1071	0.1896	48.4541	69.0672
Average	45.3477	0.1133	0.1997	55.1750	72.8114
Images	Proposed	AED	RX	LRX	CRD
ABU-b1	29.7813	0.2362	0.3361	110.3855	158.7556
ABU-b2	53.6094	0.1046	0.2219	49.9972	72.4760
ABU-b3	8.0313	0.1037	0.1683	45.2350	66.3676
ABU-b4	49.2656	0.2379	0.3042	40.6928	89.9164
Average	35.1719	0.1706	0.2576	61.5776	96.8789
Images	Proposed	AED	RX	LRX	CRD
ABU-u1	105.9844	0.1013	0.1929	54.0951	72.4728
ABU-u2	20.3281	0.1054	0.2072	59.3589	78.6753
ABU-u3	44.4063	0.1022	0.1894	46.4483	67.5966
ABU-u4	44.4531	0.1075	0.1876	58.0444	78.2669
ABU-u5	25.9063	0.1047	0.1850	57.9914	74.3002
Average	48.2156	0.1042	0.1923	55.1876	74.2624
Images	Proposed	AED	RX	LRX	CRD
SanDiego	43.9844	0.1049	0.2731	46.0064	67.3094
Images	Proposed	AED	RX	LRX	CRD
HYDICE	34.1563	0.1983	0.3281	72.2813	83.1406

by our method are 0.99706 and 0.00207 and 0.99996 and 0.00126, respectively, which are the optimal detection results compared with other four methods. The detection map obtained by the proposed method contains more anomalies. These detection results demonstrate the superior performance of the proposed SAFL model.

With the best parameters, we draw the ROC curves of the beach scene in the ABU data set and HYDICE, as shown in Fig. 11, and the area under these curves are consistent with the AUC scores reported in Tables IV and V. From Fig. 11, it is obvious that the ROC curve means that the detection result of the proposed SAFL method is always over the AED, RX, LRX, and CRD algorithms, which indicates that our method is superior to those compared methods. In particular, under the same false alarm rate, the detection accuracy of the proposed is larger than the others.

In addition, we measure the running time of each comparative method on all the testing images and the results recorded in seconds are given in Table VI. All the experiments are performed in the TensorFlow 1.3.0 and MATLAB (R2014a) environment on a server: Intel Xeon CPU E5-2650 v4 at 2.20 GHz. In comparison to the CRD and LRX methods, the proposed method is more efficient. However, the proposed method is more time-consuming than the RX and AED methods while illustrating the improvement in detection capability.

V. CONCLUSION

In this article, an unsupervised SAFL architecture that allows extracting distinctive features from the hyperspectral data is proposed. In contrast with AE and AAE, SAFL has more powerful feature representation ability with respect to high-dimensional HSIs. Based on the observation that anomalies are rare, we introduce the morphological AF to strengthen anomalies and remove the background of the fusion feature. Additionally, iteration between features and detection map is applied to reduce the false alarm rate. Specifically, the optimization is adaptively iterated based on the contribution rate of each feature that is computed by the ST matrix. Extensive experiments validate that the proposed SAFL-based anomaly detection method is superior to other methods either in detection accuracy or false alarm rate. In the future, how to implement the proposed method in real time will be the focus.

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